Jingyuan Huang

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Research Profile

PhD researcher in LLM/VLM post-training, studying **on-policy distillation (OPD)** and **multi-teacher OPD (MOPD)** for capability merging, with applications to computer-use agents. Experience in algorithm design, controlled experiments, and benchmark-based model evaluation.

Education

PhD **University of Georgia**, Computer Science — Aug 2025 - present
Advisor: Prof. Ninghao Liu

BS **The Chinese University of Hong Kong**, Artificial Intelligence — Sep 2021 - Jun 2025
Undergraduate research advisor: Prof. Michael R. Lyu

Research Experience

INF, Research Intern — May 2026 - Aug 2026
Led research design, method development, and evaluation for Trust the Right Teacher; continued collaboration on Train4Merge after the internship.

University of Georgia, Graduate Research Assistant — Aug 2025 - present
Research on OPD/MOPD and computer-use agents, co-supervised by Prof. Ninghao Liu and Prof. Xiaoming Zhai.

ARISE Lab, CUHK, Research Assistant — May 2022 - May 2025
Conducted research on LLM/VLM evaluation and benchmarking under Prof. Michael R. Lyu.

- Developed multimodal testing methods for content moderation systems (ISSTA 2023, ASE 2023) and contributed to Asclepius, a medical VLM evaluation benchmark (ACL 2025).

Selected Research

Train4Merge — First author
RL or SFT - Which Yields Better Teachers for OPD-Based Model Merging? — arXiv, Sep 2026

- Designed the study end to end, comparing **SFT- and RL-trained (GRPO) teachers** from a shared Qwen3.5-9B initialization with comparable task performance across agentic, reasoning, and GUI perception domains.
- MOPD students supervised by RL-trained teachers achieved a **60.04% three-domain mean versus 58.97%** with SFT teachers, including **23.93% versus 19.94% task success on MobileWorld**.
- Hypothesized that RL teachers' smaller parameter shifts improve OPD transfer; tested this with parameter-space diagnostics and Top-20 teacher-student prediction alignment.

Trust the Right Teacher — First author
Quality-Aware Self-Distillation for GUI Grounding — arXiv, Jun 2026

- Introduced, to our knowledge, the **first OPSD method for GUI grounding to verify teacher coordinate-token correctness** under student-generated prefixes and adapt teacher supervision accordingly.
- Designed correctness-aware supervision using **soft gating and teacher-probability scaling** in a weighted reverse-KL objective to handle unreliable teacher signals.
- Achieved **72.23% macro-average accuracy** across six GUI grounding benchmarks with Qwen3.5-9B, **2.16 percentage points above GUI-SD**.

FairLocator Co-first author
AI Sees Your Location - But With A Bias Toward The Wealthy World. Research collaboration with Prof. Jieyu Zhao, USC (May 2024 - May 2025). EMNLP 2025

- Co-developed a VLM geolocation benchmark with **1,200 images from 111 cities in 43 countries**, supporting within-country and cross-country comparisons.
- Designed and ran an evaluation pipeline for **four VLMs**, measuring geographic accuracy, geodesic error, and prediction entropy to identify economic, population, and regional biases.

Earlier Research Experience

Tencent AI Lab, Research Collaborator Oct 2022 - Sep 2023

- Designed data curation, evaluation metrics, and experiments to study cultural dominance in multilingual LLMs, resulting in an ACL 2024 publication.

Technical Skills

Programming and tools: Python, PyTorch, vLLM, Hugging Face, Git, Linux

Methods: OPD, OPSD, MOPD, SFT, GRPO, benchmark design, VLM evaluation

Additional Publications

Asclepius: A Spectrum Evaluation Benchmark for Medical Multi-Modal Large Language Models
Jie Liu, Wenxuan Wang, Yihang Su, **Jingyuan Huang**, Yudi Zhang, Cheng-Yi Li, Wenting Chen, Xiaohan Xing, Kao-Jung Chang, Linlin Shen, Michael R. Lyu | ACL 2025

Not All Countries Celebrate Thanksgiving: On the Cultural Dominance in Large Language Models
Wenxuan Wang, Wenxiang Jiao, **Jingyuan Huang**, Ruyi Dai, Jen-tse Huang, Zhaopeng Tu, Michael R. Lyu | ACL 2024

An Image is Worth a Thousand Toxic Words: A Metamorphic Testing Framework for Content Moderation Software
Wenxuan Wang, **Jingyuan Huang**, Jen-tse Huang, Chang Chen, Jiazhen Gu, Pinjia He, Michael R. Lyu | ASE 2023

Validating Multimedia Content Moderation Software via Semantic Fusion
Wenxuan Wang, **Jingyuan Huang**, Chang Chen, Jiazhen Gu, Jianping Zhang, Weibin Wu, Pinjia He, Michael R. Lyu | ISSTA 2023

Self-Supervised Scaling of Terminal Environments for Scientific Domains
Zhongzhi Li, Yucheng Shi, Zongxia Li, Junyao Yang, Ruhan Wang, Yu Wang, **Jingyuan Huang**, Jichao Yu, Ninghao Liu, Haitao Mi, Leowei Liang | arXiv, Oct 2026

Online Skill Learning for Web Agents via State-Grounded Dynamic Retrieval
Jiaxi Li, Ke Deng, Yun Wang, **Jingyuan Huang**, Yucheng Shi, Qiaoyu Tan, Jin Lu, Ninghao Liu | arXiv, Jun 2026

TRON: Targeted Rule-Verifiable Online Environments for Visual Reasoning RL
Tianze Yang, Yucheng Shi, Ruitong Sun, **Jingyuan Huang**, Ninghao Liu, Jin Sun | arXiv, Jun 2026

Towards Trustworthy GUI Agents: A Survey
Yucheng Shi, Wenhao Yu, **Jingyuan Huang**, Wenlin Yao, Wenhu Chen, Ninghao Liu | arXiv, Mar 2025; rev. Feb 2026